

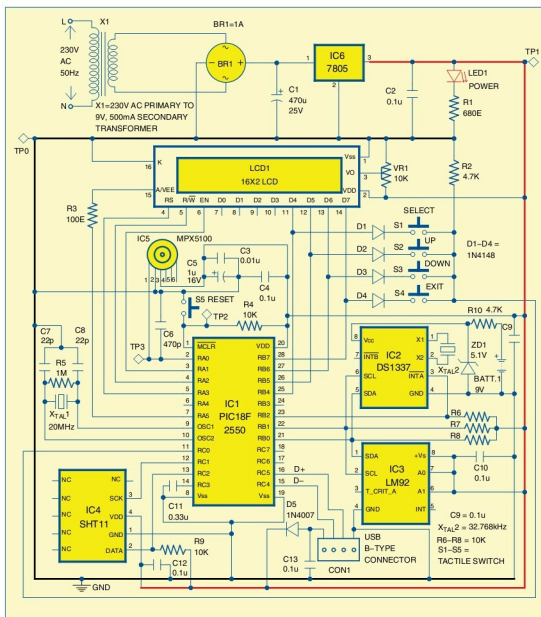


Fig. 2: Circuit diagram of a weather logger

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when S5 is pressed
TP3	Analogue output from pressure sensor

Two pins (A0, A1) are available for address selection.

Pins 1 and 2 of temperature sensor IC3 are connected with pins 21 and 22 of IC1, respectively. The address pins 6 and 7 are connected with Vcc

as shown in Fig. 2.

Humidity sensor. The SHT11 is a single-chip relative humidity and temperature sensor module comprising a calibrated digital output. The device includes a capacitive polymer-sensing element for relative humidity and a band-gap temperature sensor. Both are seamlessly coupled to a 14-bit analogue-to-digital (A/D) converter and a serial interface circuit on the same chip. The 2-wire serial interface and internal voltage regulation allow easy and fast system integration. Pins 3 and 2 of humidity sensor IC4 are con-

nected with pins 12 and 13 of IC1. Pin 2 is pulled up 'high' with resistor R9.

Pressure sensor. The MPX5100 series piezoresistive transducer is a monolithic silicon pressure sensor designed for a wide range of applications, particularly those employing a microcontroller or microprocessor with A/D inputs. This single-element transducer combines advanced micromachining techniques, thin-film metallisation and bipolar processing to provide an accurate, high-level analogue output signal that is proportional to the applied pressure.

Pin 1 of pressure sensor IC5 is connected with pin 2 of IC1. IC5 produces analogue output corresponding to pressure at this pin.

Pins 3 through 5 of IC1 are connected with EN, R/W and RS of LCD1 to control the LCD. Pins 25 through 28 of IC1 are connected with D4 through D7 data pins of LCD1. LCD1 shows the time, temperature, pressure and humidity on the screen. Pins 15 and 16 of IC1 are connected with USB connector at D- and D+ as shown in Fig. 2. Switches S1 through S4 are used to set the pressure reference and time. Switch S5 is used to reset the microcontroller.

Working of the weather logger is simple. Once powered on, the LCD shows the real-time values of the temperature, pressure and humidity on